

Sources and environmental impacts of Kura River pollution under anthropogenic pressures in Azerbaijan

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ABSTRACT

The Kura River is Azerbaijan's principal river system and supports drinking-water supply, irrigation, hydropower, fisheries and riverine ecosystems. At the same time, its transboundary character and intensive human use make it vulnerable to multiple forms of pollution. This study examines the principal anthropogenic sources of Kura River pollution and their environmental implications by integrating two independently prepared research dossiers with a targeted review of scientific and institutional literature. The original research materials were based on desk research, qualitative observation and comparison of pollution problems; no new laboratory water-quality measurements were generated. The synthesis identifies four recurrent pressure groups: untreated municipal and industrial wastewater, domestic solid waste and plastics, agricultural runoff, and transboundary pollutant loads. These pressures are linked to deterioration of water quality, habitat stress, risks to aquatic organisms, potential effects on irrigation and soil quality, and exposure pathways relevant to human health. Recent studies corroborate several of the source-level observations, including downstream changes in Kura River mineralization, historically elevated metal concentrations at transboundary sites, and substantial plastic transport from the Kura-Aras basin toward the Caspian Sea. The results indicate that pollution control cannot rely on riverbank clean-up alone. Priority measures include wastewater treatment, solid-waste prevention, agricultural nutrient management, basin-scale monitoring and stronger transboundary coordination. The study provides a concise source-pathway-impact framework that can support environmental education, local monitoring and future empirical research on the Kura River.

INTRODUCTION

The Kura River is the largest river of the South Caucasus and the main surface-water artery of Azerbaijan. Official hydrometeorological information lists a total river length of 1,515 km and a catchment area of about 188,000 km², while 906 km of the river course lies within Azerbaijan (Ministry of Ecology and Natural Resources of the Republic of Azerbaijan [MENR], 2025). The river rises in northeastern Türkiye, passes through Georgia and enters Azerbaijan before discharging into the Caspian Sea. This geographical setting gives the Kura both national and transboundary importance. Its water is used for drinking-water supply, irrigation, industrial activities and hydropower, while the river and its floodplain support fish, birds and riparian vegetation. Consequently, water quality degradation in the Kura is not a single-sector problem; it affects environmental health, food production, settlements and the wider Kura-Caspian system.

Azerbaijan's water security is strongly influenced by transboundary flows. The World Bank (2024) reports that approximately 70% of the country's surface-water resources originate outside its borders and identifies pollution, inefficient irrigation and inadequate wastewater infrastructure among the principal pressures on water security. The Kura is therefore exposed to cumulative impacts generated both upstream and within Azerbaijan. Historical research documented elevated concentrations of several metals at monitoring sites close to the Georgian and Armenian borders, demonstrating the relevance of

transboundary pollution pathways (Suleymanov et al., 2010). More recent monitoring has shown that the chemical character of the river also changes longitudinally, with substantially higher mineralization in the lower reaches than near the upstream border sections (Aliyev & Abduev, 2025). These patterns illustrate why pollution must be considered at basin scale rather than only at individual settlements.

The two original research works integrated in this paper focused on practical and visible forms of human pressure on river water. They independently emphasized untreated industrial and municipal effluents, household litter and plastics, agricultural chemicals, and environmentally harmful behaviour as major sources of concern. They also linked pollution with declining water quality, stress on fish and other aquatic organisms, possible human-health risks and reduced suitability of water for agricultural use. These themes are scientifically plausible and are consistent with the pressure categories identified in regional water-management assessments, although the source projects did not include laboratory sampling or quantified pollutant loads. The present paper therefore does not reinterpret the original observations as chemical measurements; instead, it develops them into a structured qualitative assessment and evaluates them against published evidence.

The objective of this study is to identify the main anthropogenic pollution pressures highlighted by the original research materials, connect each pressure to its principal environmental pathway and

likely receptor, and assess the extent to which recent scientific and institutional evidence supports those relationships. The study asks three questions: Which human activities emerge most consistently as sources of Kura River pollution? Through which pathways can these activities affect water quality and ecosystems? Which management responses are most directly aligned with the identified pressures? By answering these questions, the paper aims to preserve the practical focus of the original investigations while placing their findings within a more rigorous environmental management framework.

LITERATURE REVIEW

Pollution in the Kura-Aras basin has long been recognized as a transboundary environmental issue. Campana et al. (2012) described the basin as a setting in which water-quality and water-quantity problems are shared across Armenia, Azerbaijan and Georgia, with untreated urban wastewater, agricultural return flows and industrial activities contributing to cumulative pressures. The United Nations Development Programme (UNDP, 2013) similarly identified transboundary degradation as a priority issue requiring coordinated basin management. A later pressure and impact assessment prepared under the European Union Water Initiative Plus (EUWI+, 2021) classified water withdrawal, hydromorphological modification, untreated municipal and industrial discharges, and diffuse pollution from fertilized and pesticide-treated agricultural land among the main pressures affecting the lower Kura-Aras system. These assessments support the core argument of the original research works that human activities occurring at multiple points in the basin can combine to reduce ecological quality downstream.

Chemical monitoring provides more direct evidence of these pressures. Suleymanov et al. (2010) analyzed monthly samples from the Kura-Aras river system between 2004 and 2008 and found strong spatial differences in trace-metal concentrations. Sites close to the borders with Georgia and Armenia were among the most polluted, and several metals were transported substantially in suspended form. Although these data represent an earlier period, they remain important because they demonstrate that pollutant transport across national boundaries is measurable and that sediment-associated contamination can be relevant. More recent work by Aliyev and Abduyev (2025), based on monitoring at 14 points during 2020–2022, focused on major ions and mineralization rather than a broad suite of toxic contaminants. Their results showed a pronounced downstream rise in mineralization and changes in ionic composition, while also emphasizing continuing pressures from wastewater and tributary inputs. The contrast between historical metal contamination and newer ion-based monitoring also highlights the need for long-term, multi-parameter datasets rather than reliance on a single pollution indicator.

Solid waste and plastic pollution form a second major pressure pathway. This theme was prominent in the source research, where plastic bottles and household waste were identified as visible forms of river pollution and as materials that can fragment into smaller particles. Recent basin-scale studies strengthen this concern. Mousavi et al. (2025) estimated that large quantities of mismanaged plastic waste are generated across the Kura-Aras basin and modelled substantial annual plastic transport to the Caspian Sea. Ghaffari et al. (2025) subsequently applied field monitoring, unmanned aerial vehicle surveys and hydrodynamic modelling at the Kura delta, recording more than 600 debris items along a surveyed shoreline and demonstrating both local accumulation and transboundary origins of some plastic items. These findings move the plastic issue from a general visual concern to a measurable river-to-sea pollution pathway.

Agricultural pollution is more diffuse and therefore less visible than litter or wastewater outfalls. Fertilizers containing nitrogen and phosphorus can be transported through surface runoff or drainage water, increasing nutrient loads and, under suitable conditions, stimulating excessive algal production and oxygen depletion. The EUWI+ (2021) assessment identifies agricultural areas treated with fertilizers and pesticides as a diffuse pressure in the lower Kura-Aras system. The original research materials described the same mechanism in accessible terms, linking fertilizer wash-off with reduced dissolved oxygen and stress on aquatic organisms. Importantly, the present study treats this as a pressure pathway supported by environmental science rather than as a measured nutrient concentration in the authors' own field data.

The ecological consequences of river pollution can occur at several levels, from altered water chemistry to organism-level stress and habitat degradation. At the Kura River mouth, Yusifova et al. (2024) reported histological changes in fish gills and liver that were interpreted mainly as protective-adaptive responses and as evidence of some environmental stress factor, while genotoxic indicators did not exceed the reported norm. This result is important because it shows the need for cautious interpretation: ecological stress may be detectable even when a single biomarker or chemical parameter does not indicate extreme contamination. In parallel, the World Bank (2024) emphasizes the need for improved water-quality monitoring, wastewater treatment and pollution enforcement in Azerbaijan. Taken together, the literature supports a multi-pressure perspective in which point-source effluents, diffuse agricultural inputs, plastics, sediment-associated contaminants and reduced flows interact rather than act in isolation.

MATERIALS AND METHODS

This study used a qualitative integrative research design. The primary analytical corpus consisted of two independently prepared 2026 research dossiers addressing Kura River pollution and its environmental consequences. Both source works used online research and documentary review; they also incorporated qualitative observation, environmental images or maps, and simple problem-effect comparisons. One dossier emphasized the causes and broad environmental consequences of river-water pollution, while the second gave greater attention to agricultural chemicals, plastics, transboundary contamination, water-level pressures and possible mitigation measures. The original wording and thematic priorities were retained as the foundation of the analysis, but repeated material was consolidated and claims requiring empirical support were checked against external literature.

The analysis followed a three-stage coding procedure. First, statements in the two source works were grouped into pressure categories: (1) municipal and industrial wastewater, (2) domestic solid waste and plastics, (3) agricultural runoff and chemical inputs, (4) transboundary contaminant transfer, and (5) interacting water-quantity and hydromorphological pressures. Second, each pressure was linked to an environmental pathway, such as direct discharge, surface runoff, river transport, sediment transport or reduced dilution capacity. Third, likely receptors and impacts were classified as water-quality deterioration, aquatic-ecosystem stress, soil or irrigation effects, human exposure pathways, and downstream transfer to the Caspian Sea.

A targeted literature review was then used for triangulation. Priority was given to peer-reviewed articles and official reports directly concerning the Kura River, the Kura-Aras basin or Azerbaijan's water sector, with emphasis on sources published from 2010 to 2025.

Recent evidence was used where available, particularly for water-quality trends and plastic pollution. The study did not conduct chemical analysis, microbiological testing, flow measurement or statistically representative field sampling. Accordingly, no concentration, exceedance rate or causal effect is attributed to the authors' original observations unless supported by a cited external source. The resulting framework is interpretive and comparative, designed to identify convergent pressure-impact relationships and management priorities rather than to estimate pollutant loads.

RESULTS AND DISCUSSION

Dominant anthropogenic pressure pathways

The synthesis shows that wastewater discharges are the most direct point-source pressure identified across the research materials. Industrial effluents and untreated or insufficiently treated municipal wastewater can introduce organic matter, nutrients, oils, metals and other contaminants into the river. This pathway is consistent with the World Bank's (2024) assessment that Azerbaijan needs to modernize and expand wastewater treatment infrastructure, and with EUWI+ (2021), which identifies untreated discharges from riparian settlements and industrial enterprises as a major pressure. The environmental significance of point-source wastewater is amplified in river sections where flows are lower or where multiple discharges accumulate, because the river has less capacity to dilute incoming loads. The source works correctly emphasized prevention and treatment at the discharge point rather than relying only on downstream clean-up.

Domestic solid waste, particularly plastic, emerged as the most visible pollution pressure. The original materials described littering on riverbanks and plastic containers entering the water, and they connected long-term plastic persistence with risks to aquatic organisms. Recent empirical work at the Kura delta strongly supports this concern. Ghaffari et al. (2025) documented debris accumulation through standardized field monitoring and remote-sensing tools, while Mousavi et al. (2025) estimated substantial basin-scale plastic inputs to the Caspian Sea. These studies indicate that visible litter is not merely an aesthetic problem; it is part of a transport continuum in which mismanaged waste can move from settlements and riverbanks through the river network and ultimately accumulate in the delta or Caspian coastal environment.

Agricultural runoff represents a diffuse pressure that is harder to observe directly. The source research identified excessive use of nitrogen- and phosphorus-containing fertilizers as a potential cause of nutrient enrichment. This mechanism is consistent with the EUWI+ (2021) identification of agricultural land treated with fertilizers and pesticides as a diffuse source of pressure. Unlike a wastewater pipe, agricultural pollution may enter the river through many small runoff and drainage pathways, which complicates monitoring and enforcement. Management therefore requires source reduction, appropriate fertilizer application, drainage control and vegetated buffer zones rather than a single end-of-pipe treatment solution.

Transboundary transport cuts across all of these categories. The Kura receives water and pollutant loads from upstream territories before entering Azerbaijan, while internal tributaries and settlements add further pressures along the downstream course. Suleymanov et al. (2010) demonstrated strong spatial differences in trace metals and particularly high contamination at several border-related sites. Campana et al. (2012) and UNDP (2013) similarly framed the basin as a shared system in which water-quality management depends on cooperation. Therefore, an effective pollution-control strategy for

Azerbaijan must combine domestic source control with information exchange and coordinated monitoring across the basin.

Environmental consequences and receptor pathways

The environmental effects identified in the original research can be organized into three linked receptor groups. First, aquatic ecosystems are exposed through changes in chemical quality, oxygen conditions, physical habitat and ingestion or contact with pollutants. Nutrient enrichment may increase biological oxygen demand indirectly through enhanced primary production and decomposition, while plastics can create ingestion and entanglement risks and can fragment into smaller particles. Fish and benthic organisms can also be exposed to dissolved or sediment-associated metals. Yusifova et al. (2024) found tissue-level changes in fish from the Kura River mouth area that were consistent with the presence of environmental stress, although the study did not attribute those changes to a single pollutant. This supports the use of multiple indicators when evaluating ecological condition.

Second, pollution can affect human uses of river water. The source materials emphasized drinking-water safety, irrigation and agricultural productivity. These concerns should be expressed cautiously: the present study does not measure disease incidence or crop losses. Nevertheless, the pathway is clear. Poor raw-water quality can increase treatment requirements and costs, while contaminated irrigation water may transfer salts, nutrients or other pollutants to soils and agricultural systems. The strong downstream mineralization gradient reported by Aliyev and Abduev (2025), including much higher values in lower-river stations, shows that water-quality pressures are not spatially uniform. This reinforces the need to match water use with location-specific monitoring rather than assuming that all sections of the river have the same chemical status.

Third, pollution in the Kura has downstream consequences beyond the river itself. Because the Kura discharges to the Caspian Sea, river-borne pollutants and plastics can be transferred to deltaic and coastal environments. The Kura delta studies by Ghaffari et al. (2025) and Mousavi et al. (2025) provide direct evidence for the importance of this river-to-sea pathway. In management terms, this means that upstream waste prevention and wastewater treatment are also Caspian Sea protection measures. River and coastal policies should therefore be coordinated rather than treated as separate environmental domains.

Table 1. Main anthropogenic pressures, pathways and environmental implications identified in the synthesis

Pressure	Pathway and environmental implication
Municipal and industrial wastewater	Direct discharge to the river or tributaries can increase organic and chemical loading, raise treatment demand and create aquatic stress.
Solid waste and plastics	Littering and storm runoff transport plastics downstream, creating ingestion/entanglement hazards, fragmentation and transfer to the Kura delta and Caspian Sea.
Agricultural inputs	Runoff and drainage from cultivated land can transport nutrients and pesticides, contributing to diffuse pressure, possible oxygen stress and irrigation concerns.
Transboundary pollutants	Upstream river and sediment transport can create cumulative cross-border exposure to metals and other contaminants.
Reduced flow and hydromorphological change	Lower dilution, altered residence time and sediment transport can intensify water-quality pressures and ecological vulnerability.

Interaction between pollution and water-quantity pressures

Although the article focuses on pollution, the source research also noted declining water levels and inefficient water use. These issues matter because quantity and quality are interdependent. Reduced flow can increase the relative concentration of pollutants, reduce dilution capacity and alter sediment transport, while reservoirs and abstractions can change residence times and ecological conditions.

The World Bank (2024) identifies both water abstraction and pollution as major pressures, and EUWI+ (2021) reports significant hydromorphological and flow-related changes in the lower basin. The present synthesis therefore interprets pollution control as part of integrated river-basin management, not as an isolated waste-management task.

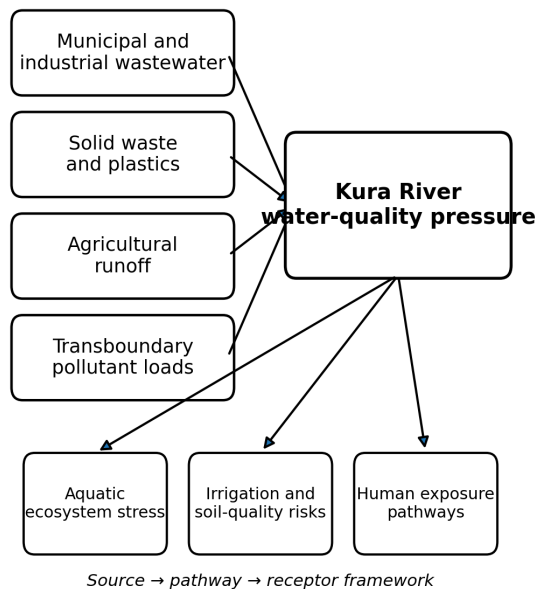


Figure 1. Source–pathway–impact framework used in the qualitative synthesis. Figure 1 summarizes the analytical logic used in the study. The framework does not assume that each pressure produces a single effect; rather, it emphasizes that several source types can converge on river water quality and can affect multiple receptors simultaneously.

Management priorities

Five management priorities emerge from the combined evidence. The first is prevention and treatment of municipal and industrial wastewater before discharge. The second is stronger solid-waste management at settlements and recreation areas along the river, combined with routine clean-up and enforcement against dumping. The third is reduction of diffuse agricultural pollution through more precise nutrient use, runoff control and riparian buffers. The fourth is a monitoring system that combines conventional chemistry with biological indicators, litter surveys and, where feasible, sediment analysis. The fifth is transboundary data sharing and coordinated basin management. These priorities directly reflect the pressures identified by the original research while also aligning with recent national and international evidence.

CONCLUSIONS

This study integrated two independent research works on Kura River pollution into a structured assessment of anthropogenic pressures, environmental pathways and management priorities. The combined analysis preserves the central observations of the original investigations: industrial and municipal wastewater, household litter and plastics, agricultural chemicals, and transboundary pollution are recurrent concerns, while the consequences extend from water quality to aquatic organisms, irrigation, human uses and the Caspian receiving environment. Scientific evidence supports the relevance of these themes. Historical monitoring documented transboundary metal contamination, recent work has shown strong downstream changes in major-ion composition and mineralization, and 2025 studies provide direct evidence that plastic waste is transported

through the Kura-Aras system toward the Caspian Sea. At the same time, the evidence also demonstrates the importance of caution. The source projects did not collect laboratory water samples, and therefore their observations should not be interpreted as quantified contamination levels or causal health estimates. Their value lies in identifying practical pressure categories that can guide more rigorous monitoring. The principal implication is that Kura River protection requires source-oriented management. Wastewater must be treated before discharge, plastics and household waste must be prevented from entering the river, agricultural runoff must be reduced, and monitoring should cover chemical, biological and solid-waste indicators. Because the river is transboundary and ultimately discharges to the Caspian Sea, local action should be connected to basin-scale cooperation. Future research should combine the qualitative pressure framework developed here with georeferenced field observations, seasonal water sampling and standardized laboratory analysis so that visible pollution pressures can be linked to measurable environmental outcomes.

AUTHOR CONTRIBUTIONS

Z.A. and J.A. contributed the original research materials, problem identification and initial environmental observations. G.G. coordinated the academic integration, methodological structuring and manuscript-level review. All authors contributed to interpretation, reviewed the final manuscript and approved the submitted version.

CONFLICTS OF INTEREST

The authors declare no conflict of interest.

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This research received no external funding.

DATA AVAILABILITY

The qualitative source materials and synthesis used in this study are described within the article. No new laboratory water-quality dataset was generated.

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